Q1 - 24 June - Shift 1

Let $A = \{z \in C : 1 \le |z-(1+i)| \le 2\}$ and $B = \{z \in A : |z-(1-i)| = 1\}$. Then, B :

(A) is an empty set

(B) contains exactly two elements

(C) contains exactly three elements

(D) is an infinite set

Space for your notes:

Q2 - 24 June - Shift 2

Let S = $\{z \in \mathbb{C} : |z-3| \le 1 \text{ and } z(4+3i) + \bar{z}(4-3i) \le 24\}$.

If $\alpha + i\beta$ is the point in S which is closest to 4i, then $25(\alpha + \beta)$ is equal to ______.

Space for your notes:

Q3 - 25 June - Shift 1

Let a circle C in complex plane pass tltrough the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg(z) is equal to :

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$ (B) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$

(C) $\tan^{-1}(3) - \pi$ (D) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

Space for your notes:

Q4 - 25 June - Shift 2

Let z_1 and z_2 be two complex numbers such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$. Then

(A) $\arg z_2 = \frac{\pi}{4}$ (B) $\arg z_2 = -\frac{3\pi}{4}$

(C) $\arg z_1 = \frac{\pi}{4}$ (D) $\arg z_1 = -\frac{3\pi}{4}$

Space for your notes:

Q5 - 26 June - Shift 1

Let $A = \left\{z \in C : \left|\frac{z+1}{z-1}\right| < 1\right\}$

and $B = \left\{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$.

Then $A \cap B$ is :

(A) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(B) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

(C) an empty set

(D) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

Space for your notes:

Q6 - 26 June - Shift 2

If $z^2 + z + 1 = 0$, $z \in \mathbb{C}$, then $\left|\sum_{n=1}^{15}\left(Z^n + (-1)^n \frac{1}{Z^n}\right)^2\right|$ is equal to ______ .

Space for your notes:

Q7 - 27 June - Shift 1

The area of the polygon, whose vertices are the non-real roots of the equation $\bar{z} = iz^2$ is :

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$

(C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Space for your notes:

Q8 - 27 June - Shift 2

The number of points of intersection of $|z-(4+3i)| = 2$ and $|z| + |z-4| = 6$, $z \in C$ is :

(A) 0 (B) 1

(C) 2 (D) 3

Space for your notes:

Q9 - 28 June - Shift 1

The number of elements in the set

$\{z = a + ib \in \mathbb{C} : a, b \in \mathbb{Z} \text{ and } 1 < |z - 3 + 2i| < 4\}$ is

______.

Space for your notes:

Q10 - 28 June - Shift 2

Sum of squares of modulus of all the complex numbers z satisfying $\bar{z} = iz^2 + z^2 - z$ is equal to

Space for your notes:

Q11 - 29 June - Shift 1

Let α and β be the roots of the equation $x^2 + (2i - 1) = 0$. Then, the value of $|\alpha^8 + \beta^8|$ is equal to :

(A) 50 (B) 250

(C) 1250 (D) 1500

Space for your notes:

Q12 - 29 June - Shift 1

Let $S = \{z \in \mathbb{C} : |z - 2| \leq 1, z(1 + i) + \bar{z}(1 - i) \leq 2\}$. Let $|z - 4i|$ attains minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. If $5(|z_1|^2 + |z_2|^2) = \alpha + \beta\sqrt{5}$, where α and β are integers, then the value of $\alpha + \beta$ is equal to _____.

Space for your notes:

Q13 - 29 June - Shift 2

Let arg (z) represent the principal argument of the complex number z. The, $|z| = 3$ and $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}$ intersect:

(A) Exactly at one point

(B) Exactly at two points

(C) Nowhere

(D) At infinitely many points.

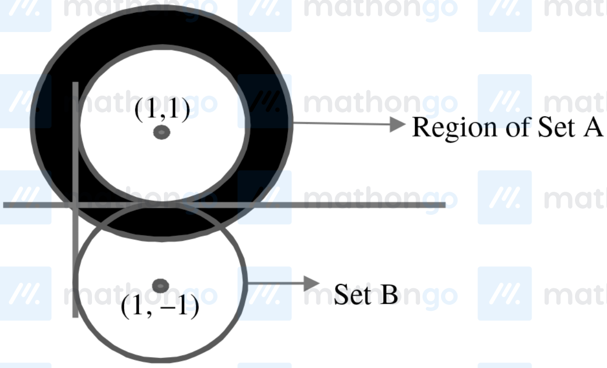
Space for your notes:

Answer Key

Q1 (D)	**Q2** (80)	**Q3** (B)	**Q4** (C)
Q5 (B)	**Q6** (2)	**Q7** (A)	**Q8** (C)
Q9 (40)	**Q10** (2)	**Q11** (A)	**Q12** (26)
Q13 (C)			

Q1 (D)

$A = \{z \in C : 1 \le |z-(1+i)| \le 2\}$



$B = \{z \in A : |z-(1-i)| = 1\}$.

$A \cap B$ has infinite set.

Q2 (80)

$|z-3| \le 1$

represent pt. i/s circle of radius 1 & centred at (3, 0)

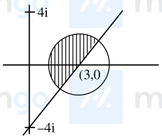
$z(4+3i)+\bar{z}(4-3i) \le 24$

$(x+iy)(4+3i)+(x-iy)(4-3i) \le 24$

$4x+3xi+4iy-3y+4x-3ix-4iy-3y \le 24$

$8x-6y \le 24$

$4x-3y \le 12$



minimum of (0, 4) from circle $= \sqrt{3^2+4^2}-1=4$

will lie along line joining (0, 4) & (3, 0)

$\therefore$ equation line

$\frac{x}{3}+\frac{y}{4}=1 \Rightarrow 4x+3y=12$ … (i)

equation circle $(x-3)^2+y^2=1$ … (ii)

$\left(\frac{12-3y}{4}-3\right)^2+y^2=1$

$\left(\frac{-3y}{4}\right)^2+y^2=1$

$\frac{25y^2}{16}=1 \Rightarrow y=\pm\frac{4}{5}$

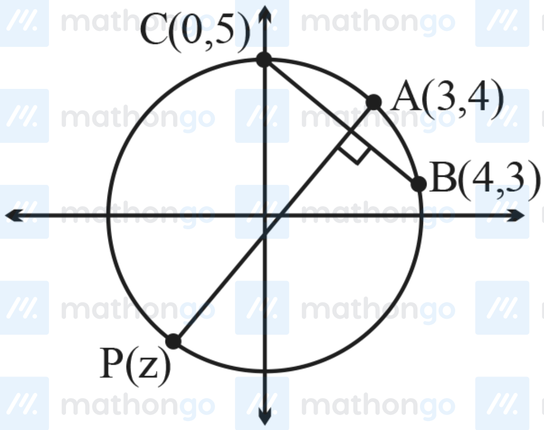
for minimum distance $y=\frac{4}{5}$

$\therefore x=\frac{12}{5}$

$\therefore 25(\alpha+\beta)=25\left(\frac{4}{5}+\frac{12}{5}\right)$

$=16\times 5=80$

Q3 (B)



Slope of BC $= \frac{3-5}{4-0} = -\frac{1}{2}$

Slope of AP $= 2$
equation of AP : $y - 4 = 2(x - 3)$
$\Rightarrow y = 2(x - 1)$
P lies on circle $x^2 + y^2 = 25$
$\Rightarrow x^2 + (2(x - 1))^2 = 25$

$\Rightarrow x = -\frac{7}{5}$ and $y = -\frac{24}{5}$

$\Rightarrow \arg(z) = \tan^{-1}\left(\frac{24}{7}\right) - \pi$

Q4 (C)

$\overline{z}_1 = i\overline{z}_2$

$z_1 = -iz_2$

$\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$

$\arg\left(-i\frac{z_2}{\overline{z}_2}\right) = \pi \qquad \arg(z_2) = \theta$

$-\frac{\pi}{2} + \theta + \theta = \pi$

$2\theta = \frac{3\pi}{2}$

$\arg(z_2) = \theta = \frac{3\pi}{4}$, $\arg z_1 = \frac{\pi}{4}$

Q5 (B)

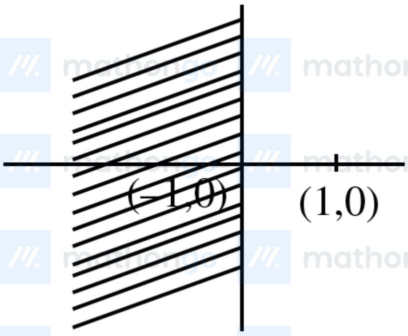
Set A

$$\Rightarrow \left|\frac{z+1}{z-1}\right|<1$$

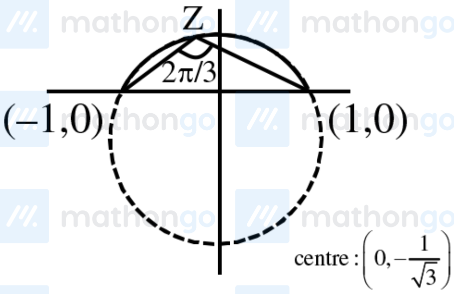
$$\Rightarrow |z+1|<|z-1|$$

$$\Rightarrow (x+1)^2+y^2<(x-1)^2+y^2$$

$$\Rightarrow x<0$$



Set B



$$\Rightarrow \arg\left(\frac{z-1}{z+1}\right)=\frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-1}\right)-\tan^{-1}\left(\frac{y}{x+1}\right)=\frac{2\pi}{3}$$

$$\Rightarrow x^2+y^2+\frac{2y}{\sqrt{3}}-1=0$$

$A \cap B$

$$\Rightarrow \text{Centre}\left(0,-\frac{1}{\sqrt{3}}\right)$$

Q6 (2)

$z^2 + z + 1 = 0 \Rightarrow z = w, w^2$

$$\left|\sum_{n=1}^{15}\left(z^n + (-1)\frac{1}{z^n}\right)^2\right| = \left|\sum_{n=1}^{15}\left(z^{2n} + \frac{1}{z^{2n}} + 2(-1)^n\right)\right|$$

$$= \left|\sum_{n=1}^{15} w^{2n} + \frac{1}{w^{2n}} + 2(-1)^n\right|$$

$$= \left|\frac{w^2\left(1-w^{30}\right)}{1-w^2} + \frac{\frac{1}{w^2}\left(1-\frac{1}{w^{30}}\right)}{1-\frac{1}{w^2}} + 2(-1)\right|$$

$$= \left|\frac{w^2(1-1)}{1-w^2} + \frac{\frac{1}{w^2}(1-1)}{1-\frac{1}{w^2}} - 2\right|$$

$= |0 + 0 - 2| = 2$

Q7 (A)

$\Rightarrow$ Let z = x + iy, x, y $\in$ R

Now $\bar{z} = iz^2$

then $x - iy = i(x^2 - y^2 + 2xyi)$

$x - iy = i(x^2 - y^2) - 2xy$

$\Rightarrow x = -2xy \ \& \ -y = x^2 - y^2$

$\Rightarrow x(1 + 2y) = 0$

$x = 0$ or $y = -\frac{1}{2}$

Put $x = 0$ in $-y = x^2 - y^2$

We get $y = y^2$

$\Rightarrow y = 0, 1$

Similarly

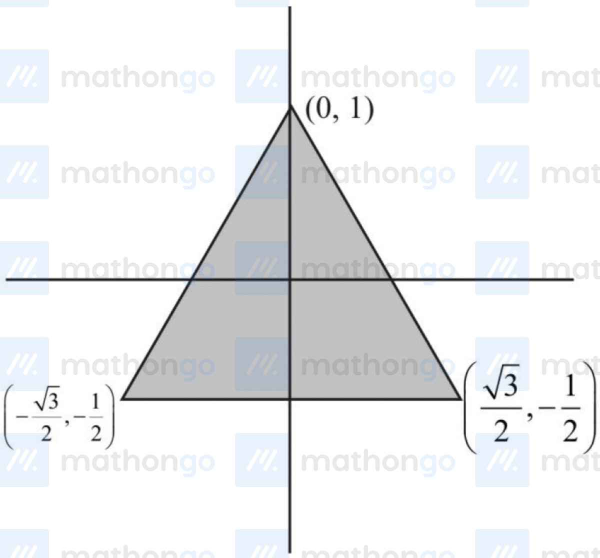
Put $y = -\frac{1}{2}$ $in - y = x^2 - y^2$

$\Rightarrow \frac{1}{2} = x^2 - \frac{1}{4}$

$\Rightarrow x^2 = \frac{3}{4}$

$x = \pm\frac{\sqrt{3}}{2}$

$z = \left(0, i, \frac{\sqrt{3}}{2} - \frac{1}{2}i, -\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$



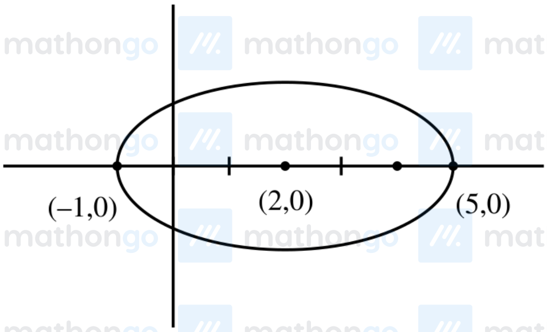
$Area = \frac{1}{2}.\left(\sqrt{3}\right)\left(\frac{3}{2}\right)$

$= \frac{3\sqrt{3}}{4}$

Q8 (C)

$C : (x-4)^2 + (y-3)^2 = 4$

$E : \frac{(x-2)^2}{9} + \frac{y^2}{5} = 1$



Lower Extremity of vertical diameter of

circle $\rightarrow (4,1)$

Put in ellipse $\Rightarrow \frac{(4-2)^2}{9} + \frac{1}{5} - 1$

$= \frac{4}{9} + \frac{1}{5} - 1$

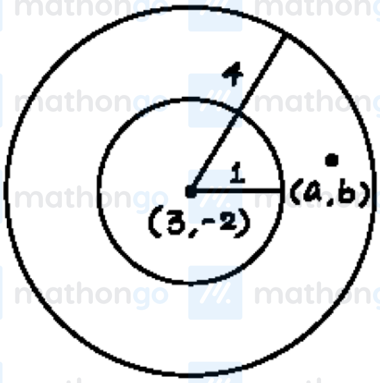
$= \frac{29}{45} - 1 < 0$

Two Solutions

Answer (C)

Q9 (40)

$1 < |Z - 3 + 2i| < 4$



$1 < (a-3)^2 + (b+2)^2 < 16$

$(0,\pm 2),(\pm 2,0),(\pm 1,\pm 2),(\pm 2,\pm 1)$

$(\pm 2,\pm 3),(3\pm,\pm 2),(\pm 1,\pm 1),(2\pm,\pm 2)$

$(\pm 3,0),(0,\pm 3),(\pm 3 \pm 1),(\pm 1,\pm 3)$

Total 40 points

Q10 (2)

$z + \bar{z} = iz^2 + z^2$

Consider z = x + iy

$2x = (i + 1)(x^2 - y^2 + 2xyi)$

$\Rightarrow 2x = x^2 - y^2 - 2xy$ and $x^2 - y^2 + 2xy = 0$

$\Rightarrow 2x = -4xy$

$\Rightarrow x = 0$ or $y = \frac{-1}{2}$

Case 1 : $x = 0 \Rightarrow y = 0$ here $z = 0$

Case 2 : $y = \frac{-1}{2}$

$\Rightarrow 4x^2 - 4x - 1 = 0$

$(2x-1)^2 = 2$

$2x - 1 = \pm\sqrt{2}$

$x = \frac{1 \pm \sqrt{2}}{2}$

Here $z = \frac{1+\sqrt{2}}{2} - \frac{i}{2}$ or $z = \frac{1-\sqrt{2}}{2} - \frac{i}{2}$

Sum of squares of modulus of z

$= 0 + \frac{(1+\sqrt{2})^2 + 1}{4} + \frac{(1-\sqrt{2})^2 + 1}{4} = \frac{8}{4} = 2$

Q11 (A)

$X^2 = 1 - 2i \quad \Rightarrow \alpha^2 = 1 - 2i, \quad \beta^2 = 1 - 2i$

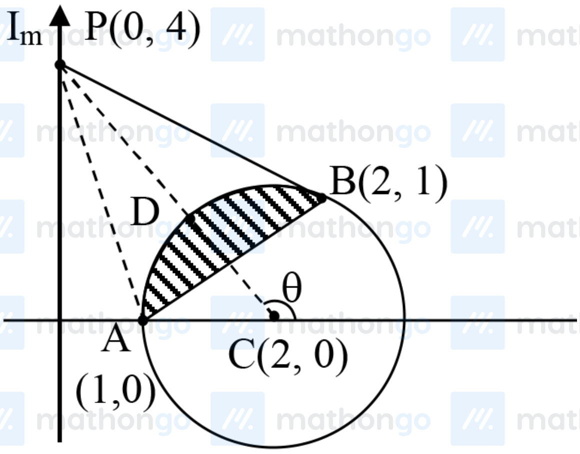
Hence $\alpha^8 = \beta^8$

$|\alpha^8 + \beta^8| = |2\alpha^8| = 2|\alpha^2|^4$

$= 2\sqrt{5}^4 = 50$

Q12 (26)

$|z-2| \le 1$



$(x-2)^2 + y^2 \le 1$ ….. (1)

&

$z(1+i) + \bar{z}(1-i) \le 2$

Put $z = x + iy$

$\therefore\ x - y \le 1$ ….. (2)

$PA = \sqrt{17}$, $PB = \sqrt{13}$

Maximum is PA & Minimum is PD

Let D $(2 + \cos\theta, 0 + \sin\theta)$

$\therefore\ m_{cp} = \tan\theta = -2$

$\cos\theta = -\frac{1}{\sqrt{5}}$, $\sin\theta = \frac{2}{\sqrt{5}}$

$\therefore\ D\left(2 - \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

$\Rightarrow\ z_1 = \left(2 - \frac{1}{\sqrt{5}}\right) + \frac{2i}{\sqrt{5}}$

$|z_1| = \frac{25 - 4\sqrt{5}}{5}$ & $z_2 = 1$

$\therefore\ |z_2|^2 = 1$

$\therefore\ 5\left(|z_1|^2 + |z_2|^2\right) = 30 - 4\sqrt{5}$

$\therefore\ \alpha = 30$

$\beta = -4$

$\therefore\ \alpha + \beta = 26$

Q13 (C)